

Analytical Jacobian for steady non-linear heat flow

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The steady-state temperature profile of the ice shell is therefore determined by the following non-linear heat conduction problem:

PDE: $-\nabla \cdot [\kappa(T)\nabla T] = 0$ on $z \in [0, H]$

BCs: $T(0) = T_o$ and $T(H) = T_s$

Here the thickness of the ice shell is $H = 30$ km, the ocean and surface temperatures are $T_o = 273$ K and $T_s = 100$ K, respectively. The dependence of the thermal conductivity on the temperature is $\kappa = \kappa_0/T$, where $\kappa_0 = 612$.

In 1D this problem has the following analytic solution:

$$T(z) = T_o e^{\ln(T_s/T_o)z/H}$$

$$q(z) = -\frac{\kappa_0}{H} \ln(T_s/T_o) = \text{constant}.$$

For the parameters given this leads to an increase of the thermal conductivity from 2.24 at the base to 6.12 W/(m K) at the top (a factor of 2.7).

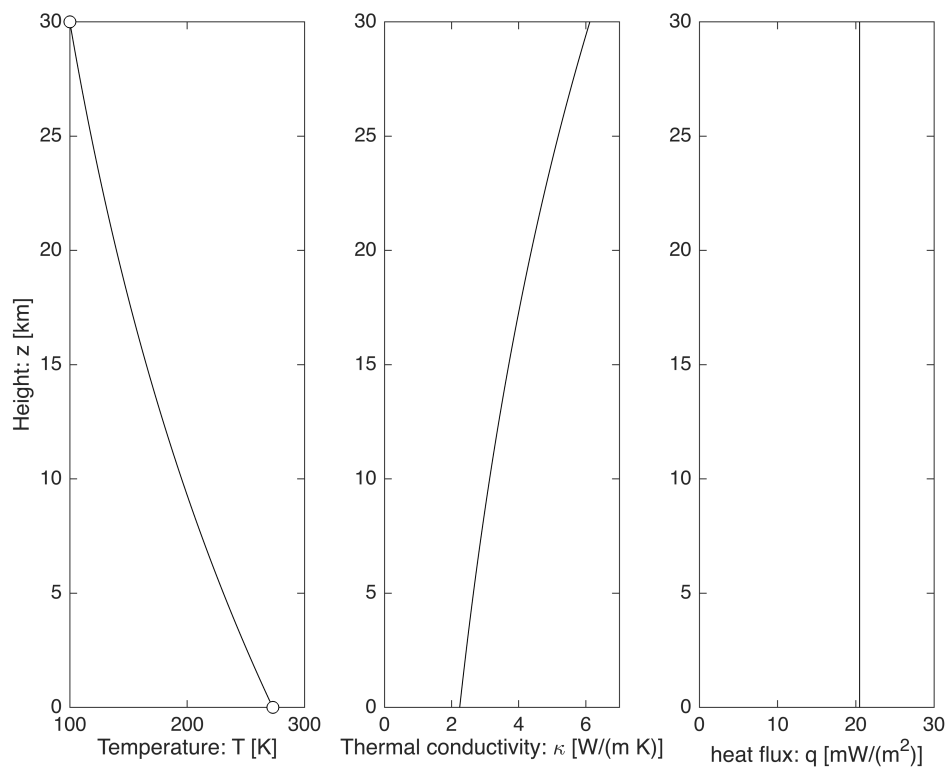
```
kappa0 = 612;
kappa = @(T) kappa0./T;
H = 30e3; % [m] ice shell thickness
Ts = 100; % [K] surface temperature
To = 273; % [K] ocean temperature
z_ana = linspace(0,H,1e2);

T_ana = @(z) To*exp(log(Ts/To)*z/H);
q_ana = @(z) -kappa0/H*log(Ts/To)+0*z;

subplot 131
plot(T_ana(z_ana),z_ana/1e3,'k-'), hold on
plot([To Ts],[0 H]/1e3,'ko','MarkerFaceColor','w')
xlabel('Temperature: T [K]')
ylabel('Height: z [km]')

subplot 132
plot(kappa(T_ana(z_ana)),z_ana/1e3,'k-')
xlabel('Thermal conductivity: \kappa [W/(m K)]')
xlim([0 7])

subplot 133
plot(q_ana(T_ana(z_ana))*1e3,z_ana/1e3,'k-'), hold on
xlabel('heat flux: q [mW/(m^2)]')
xlim([0 30])
```



Numerical Jacobian

To form the analytic Jacobian of the discrete system we need a grid

```
Grid.xim = 0; Grid.xmax = H; Grid.Nx = 2e1;
Grid = build_grid(Grid);
[D,G,~,I,M] = build_ops(Grid);
fs = spalloc(Grid.N,1,0);
```

and the derivative of the non-linear thermal conductivity

$$\frac{d\kappa}{dT} = -\frac{\kappa_0}{T^2}$$

```
dkappa = @(u) -kappa0./(u.^2);
```

The residual is given by

$$\mathbf{r}(\mathbf{u}) = -\mathbf{D}[\mathbf{K}d(\mathbf{u}) \mathbf{G} \mathbf{u}] - \mathbf{f}_s$$

```
Kd = @(u) comp_mean(kappa(u),M,1,Grid,1); % arithmetic average
res = @(u) -D*Kd(u)*G*u - fs;
```

The the numerical Jacobian is computed as follows

```
function [Jac] = comp_jacobian_steady_heat_flow_num(res,u,Grid,epsilon)
```

```

u_perturb=u;
% Pre-allocate storage (important to speed up the for loop)
Jac = spalloc(Grid.N,Grid.N,3*Grid.N);

% %% Computing the Jacobian by finite difference
% % Loop over each unknown and
% % 1) Perturb it
% % 2) Compute the change in residual vector
% % 3) Store the change from unperturbed vector as column of Jacobian
% % 4) Reset the perturbed value

for i=1:Grid.N
    u_perturb(i)=u(i)+epsilon;
    Jac(:,i)=(res(u_perturb)-res(u))/epsilon;
    u_perturb(i)=u(i);
end
end

```

Solving the non-linear steady heat conduction with Newton-Raphson method

```

BC.dof_dir = [Grid.dof_xmin;Grid.dof_xmax];
BC.dof_f_dir = [Grid.dof_f_xmin;Grid.dof_f_xmax];
BC.g = T_ana(Grid.xc(BC.dof_dir));

[B,N,fn,BC] = build_bnd(BC,Grid,I);

```

Newton iteration

```

eps_jac = 1e-6;      % perturbation for finite difference
tol = 1e-6;          % convergence tolerance
nmax = 10;           % maximum number of iterations

u = (To+Ts)/2*ones(Grid.N,1);

% BC in NR are added to initial guess -> homogeneous during iteration
u(BC.dof_dir) = BC.g;
BC.g = [0;0];

nres = 1; ndu = 1; n = 0;
while (nres > tol || ndu > tol) && n < nmax
    % 1) Solve for update
    Jac = comp_jacobian_steady_heat_flow_num(res,u,Grid,eps_jac);
    du = solve_lbvp(Jac,-res(u),B,BC.g,N);
    u = u+du;

    % 2) check for convergence
    nres = norm(N'*res(u));
    ndu = norm(du);
end

```

```

n = n+1;
fprintf('it = %d: nres = %3.2e ndu = %3.2e\n',n,nres,ndu)
end

```

```

it = 1: nres = 5.08e-05 ndu = 2.07e+02
it = 2: nres = 3.66e-06 ndu = 2.44e+01
it = 3: nres = 1.94e-08 ndu = 1.21e+00
it = 4: nres = 4.37e-13 ndu = 4.41e-03
it = 5: nres = 2.82e-19 ndu = 8.03e-08

```

```

% Compute heat flux
q = comp_flux(D,Kd(u),G,u,fs,Grid,BC);

figure
subplot 131
plot(T_ana(z_ana),z_ana/1e3,'k-'), hold on
plot(u,Grid.xc/1e3,'ro','MarkerFaceColor','w')
plot([To Ts],[0 H]/1e3,'ko','MarkerFaceColor','w')
xlabel('T [K]')
ylabel('Height: z [km]')

subplot 132
plot(kappa(T_ana(z_ana)),z_ana/1e3,'k-'), hold on
plot(kappa(u),Grid.xc/1e3,'ro','MarkerFaceColor','w')
xlabel('\kappa [W/(m K)]')
xlim([0 7])

subplot 133
plot(q_ana(T_ana(z_ana))*1e3,z_ana/1e3,'k-'), hold on
plot(q*1e3,Grid.xf/1e3,'ro','MarkerFaceColor','w')
xlabel('q [mW/(m^2)]')
xlim([0 30])

```

